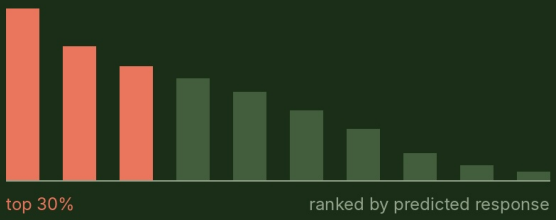
Experimentation
Uplift modeling
Decision engines



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Who Should Get the Coupon?

Who a Coupon Moves, and How Big It Should Be: A Simulated Uplift Study

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ABSTRACT

A coupon only pays off if it changes what a customer does; one used by someone who would have ordered anyway is a discount given away. This study asks who a coupon actually moves, and how big it should be. In a simulated randomized test of 15,000 customers, a $5-off-$30 coupon lifted 30-day conversion from 5.5% to 12.6%, yet about 44% of redemptions would have happened without it. An uplift model, checked on held-out customers, ranked customers by how much the coupon moves them. A second simulated test chose the coupon amount per customer: personalized amounts beat the best single offer by 11% with 31% fewer coupons, while blanket $8 coupons lost money. In both tests, a simple rule by customer segment did about as well as the model.

Keywords: uplift modeling · incremental lift · randomized experiment · customer segmentation · promotion analytics · marketing ROI

+7.0 pp
conversion lift from the offer (95% CI 6.1 to 7.9)

44%
of redemptions would have happened without the offer

+12.0 pp
lift in the model-ranked top 30% of customers

58%
of net value on 34% of the discount spend

01 Introduction

Imagine a business deciding which customers get a $5-off-$30 coupon. Promotions are among the most common growth levers and among the easiest to waste. A coupon redeemed by a customer who was already going to buy is a margin giveaway, not a growth event. The quantity that matters is therefore not who responds to an offer but who responds *because of it*.

This project builds an end-to-end workflow around that question: (i) estimate the average treatment effect from a randomized test, (ii) decompose it by lifecycle segment, (iii) rank customers by estimated incremental response, and (iv) convert the ranking into offer decisions under a fixed budget. The data are simulated (Section 2), so the numbers describe a scenario, not a real platform. The workflow is the deliverable: a platform could run it on its own experiment data, margins and budget to estimate the revenue impact of personalized coupons before rolling them out.

02 Data and Experimental Design

The dataset contains 15,000 synthetic customers described by twelve pre-experiment features: tenure, acquisition channel, geography, order counts over 30, previous 30, and 90 days, lifetime orders and spend, average order value, recency, and frequency trend.

A behavioral rule set places each customer in a lifecycle segment: Active (5,882), Lapsed (4,109), New (3,177), or At Risk (1,832). Customers were then randomly assigned to control (n = 7,409; no offer) or treatment (n = 7,591; **$5 off a $30 minimum purchase**). The outcome is conversion, a purchase within 30 days of assignment, with order value recorded for the economics.

Scope note. Segment-level treatment effects were assigned when the data were generated, so the results describe a simulated scenario and test whether the pipeline recovers a known structure, not how real customers behave. Each step is auditable against that known answer.

1. **Synthetic customers**
   15,000 customers · 12 features · 4 lifecycle segments
2. **Randomized experiment**
   Control vs. $5 off $30 · 30-day conversion
3. **Segment analysis**
   Lift, wasted offers, and net value by segment
4. **Uplift scoring**
   Two Random Forests · p(treatment) − p(control)
5. **Held-out validation**
   Decile lift · Qini curve · targeting policies
6. **Decision engine**
   Budget-constrained offers · CSV in, CSV out

Figure 1. End-to-end workflow, from synthetic data to a deployable decision tool.

03 Experiment Results

	Control	Treatment
Customers	7,409	7,591
Converted	410	953
Conversion rate	5.53%	12.55%
Absolute lift	+7.02 pp (95% CI 6.11 to 7.93)	
Relative lift		+127%
Incremental conversions		533 of 953

Table 1. Experiment outcome (n = 15,000). Lift is the treatment minus control conversion rate, in percentage points.

The offer more than doubled conversion (z = 15.1, p < 0.001). Of 953 treated conversions, an estimated 533 were incremental. The remaining 420 (44%) would have occurred anyway, so nearly half of all redeemed discounts subsidized purchases that needed no incentive. Incremental revenue was $39,825 against $4,765 in discounts, a net of $35,060 (revenue-based, not margin), or $8.94 of discount per incremental conversion.

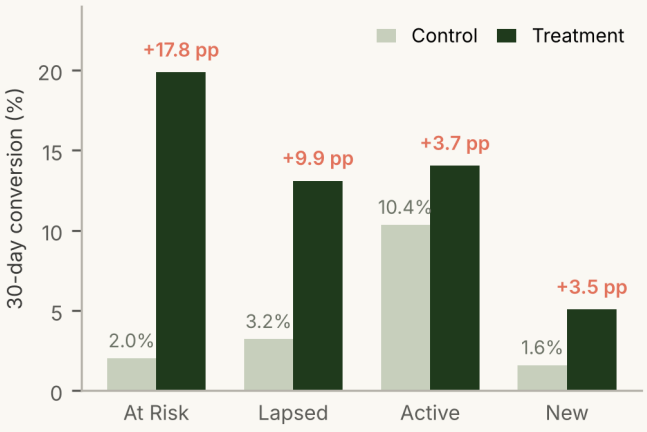


Figure 2. Conversion by lifecycle segment. Labels show the lift from treatment.

Segment	Cust.	Ctrl	Treat	Lift (pp)	Wasted*	Net value
At Risk	1,832	2.0%	19.9%	+17.8	10%	$11.0K
Lapsed	4,109	3.2%	13.1%	+9.9	25%	$14.2K
Active	5,882	10.4%	14.0%	+3.7	74%	$5.9K
New	3,177	1.6%	5.1%	+3.5	32%	$3.9K

*Table 2. Segment results. *Wasted: share of treated conversions that would have occurred without the offer. Net value is revenue-based, after discount cost.*

At Risk customers responded most (2.0% to 19.9%), and only 10% of their redemptions were wasted. Active customers were the opposite: a 10.4% baseline conversion rate meant 74% of redeemed offers went to people who were already buying. Lapsed customers, though less responsive per head, produced the largest total net value ($14.2K) because of their volume, a reminder that rate and scale answer different questions.

04 Predicting Who Responds

Two Random Forest classifiers (300 trees, depth up to 8, at least 20 samples per leaf) estimate conversion probability under control and under treatment from the twelve features. Their difference, *p*(treatment) − *p*(control), is each customer's estimated incremental response, the two-model approach to uplift. Each arm was split 75/25 with stratification, and all evidence below comes from the held-out quarter (n = 3,751).

Discrimination was modest: held-out ROC-AUC was 0.71 for the control model and 0.61 for the treatment model. Uplift is a harder target than conversion, so the better test is whether the ranking separates responders. It does. Across deciles of predicted response, observed lift falls from +15.2 pp in the top decile to +1.6 pp in the bottom (Spearman correlation −0.89; Figure 3). The top decile is 92% At Risk and 6% Lapsed.

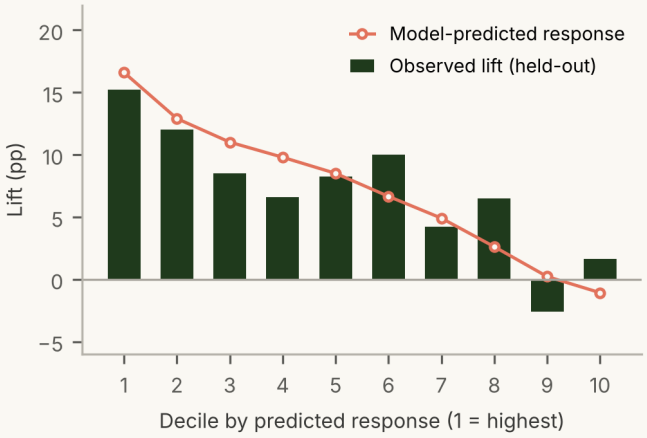


Figure 3. Predicted response versus observed lift by decile on held-out customers (about 375 per decile).

The models are well calibrated at the base rate (mean predicted control probability 5.4% against 5.6% observed), and top-decile predicted lift (+16.6 pp) is close to the observed +15.2 pp. Mid-ranked deciles are noisy at about 375 customers each. An earlier version used class weighting, which inflated predicted probabilities to 37% and was rejected.

05 Targeting Economics

Targeting the top 30% of customers by estimated response produced an observed lift of +12.0 pp (95% bootstrap CI 8.9 to 15.1), against +4.8 pp (2.5 to 6.9) in the remaining 70%. The 7.1 pp gap is itself significant (CI 3.5 to 10.8).

Policy	Cust.	Lift (pp)	Net value	$ / cust.	Return/$1
Top 10%	376	+15.2	$2,247	$5.98	15.5x
Top 20%	751	+13.6	$3,853	$5.13	14.0x
Top 30%	**1,126**	**+12.0**	**$4,849**	**$4.31**	**12.0x**
Top 50%	1,876	+10.1	$6,131	$3.27	9.7x
Top 70%	2,626	+9.3	$7,931	$3.02	8.8x
Everyone	3,751	+7.0	$8,400	$2.24	7.1x
Random 30%	1,125	+7.0	$2,520	$2.24	7.1x
Segment rule	1,522	+11.9	$6,428	$4.22	11.1x
Model, same reach	1,522	+10.6	$5,629	$3.70	10.6x

Table 3. Held-out targeting policies. Random rows are expected values at equal reach; segment rule = At Risk + Lapsed. Revenue-based.

That policy captured 68 of the 132 incremental conversions (51%) that blanket targeting produces, 58% of net value, and 34% of the discount cost, lifting net value per targeted customer from $2.24 to $4.31. Against random targeting at the same reach the model roughly doubles value per customer. Against a simple lifecycle rule the picture is less flattering: targeting At Risk and Lapsed customers (1,522) earned $6,428 net, while the model's top 1,522 earned $5,629. In this synthetic data the segment explains most of the response because the generator assigns lift by segment, leaving the model little to find beyond it.

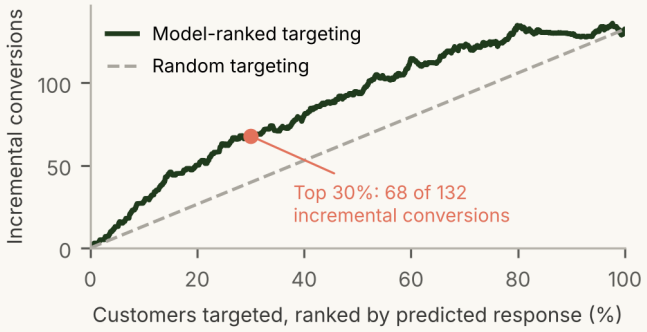


Figure 4. Cumulative incremental conversions on held-out customers, ranked by model versus random order.

06 Choosing the Coupon Amount

A single offer answers who gets a coupon, not how big it should be. To test that, a second simulated experiment randomly gave 40,000 customers no coupon or $3, $5 or $8 off a $30 order. One model per arm predicts each customer's chance to order under every option. Each customer then gets the amount with the highest expected net value, (p(order | coupon) − p(order | none)) × V − p(order | coupon) × amount, or no coupon if none is positive. V, what the platform earns per order, is set to $12 as a scenario assumption.

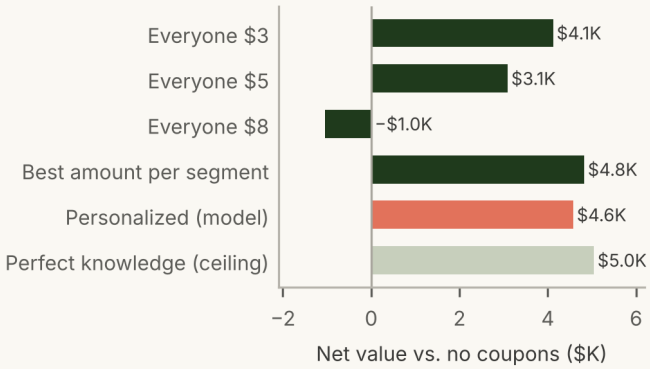


Figure 5. Policies scored on 12,000 held-out customers against the simulator's true response. Coral: model-personalized amounts.

Sending everyone $8 lost money: the extra orders did not cover discounts paid to customers who would have ordered anyway. Personalized amounts earned $4,578, 11% more than the best single offer for everyone ($3, $4,121), while sending 31% fewer coupons. The model withheld coupons from 77% of Active customers and gave At Risk and Lapsed customers mostly $3 or $5. Again the simple baseline held up: one amount per segment, chosen from training data, earned $4,827, against a ceiling of $5,046 with perfect knowledge. Estimates from observed outcomes alone (inverse-propensity weighting) ranked the policies the same way.

Policy	Sent	Cost	Net value	IPW est.
Everyone $3	12,000	$4,309	$4,121	$3,495
Everyone $5	12,000	$8,500	$3,093	$2,579
Everyone $8	12,000	$15,528	−$1,044	−$1,433
Best amount per segment	7,276	$2,911	$4,827	$5,179
Personalized (model)	**8,257**	**$4,506**	**$4,578**	**$4,747**
Perfect knowledge	7,729	$3,791	$5,046	$5,247

Table 4. Coupon policies on 12,000 held-out customers. Sent = coupons sent; cost = expected discount spend. Net value uses the simulator's true response; IPW est. uses observed outcomes only. $12 per order.

Segment	No coupon	$3 off	$5 off	$8 off
At Risk	0%	51%	49%	0%
Lapsed	0%	58%	37%	4%
New	4%	33%	56%	6%
Active	77%	16%	7%	0%

Table 5. Coupon amounts chosen by the personalized model, by segment (share of held-out customers).

07 Decision Engine

The scoring logic is packaged as a command-line tool (Python, scikit-learn). Given a CSV of customers it scores both outcomes, estimates incremental revenue and offer cost, screens eligibility, and selects customers by net value until the budget is spent, with a reason for each decision.

On a 200-customer batch with a $25 budget, 117 customers passed the eligibility screen and 29 received offers (Lapsed 11, At Risk 12, New 5, Active 1), using $24.58.

08 Recommendations

- **Size coupons by segment:** no coupon for Active customers, $3 to $5 for At Risk and Lapsed. Blanket $8 coupons lost money.
- **Start with the simple rule:** target At Risk and Lapsed customers. It matched the model here, and they are 83% of the model's own top 30%.
- **Hold back discounts from Active, high-frequency customers.** Roughly three in four redemptions there are subsidies; use non-price levers instead.
- **Measure incremental net value per discount dollar,** not redemption rate.
- **Test before scaling:** segment rule against model-ranked and random holdouts, on a real experiment.

09 Limitations and Next Steps

- **Synthetic data.** Effects were built into the generator; validation on a real experiment is the essential next step.
- **Platform value per order.** The $12 per order is an assumption; the best amounts shift with it and with real margins.
- **Simple uplift method.** The two-model approach is a baseline. Compare against X-learners, causal forests, and uplift trees, with Qini confidence intervals.
- **Economics.** Value is revenue-based and budget selection is greedy; use contribution margin and an explicit optimizer.
- **Single split.** Held-out results use one 75/25 split, about 3,750 customers.

10 Conclusion

Nearly half the discounts went to customers who needed none, and a simple segment rule captured most of the value. Run on a platform's own experiment data with real margins, this workflow estimates the revenue impact of personalized coupons before they ship.